Intelligent Log Files Analyzer

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ABSTRACT

This project presents a web-based Intelligent Log File Analyzer that uses a BERT-based model to automatically analyze system logs, detect anomalies, predict failures, and identify security threats — delivering actionable insights through interactive dashboards to strengthen system reliability and cybersecurity posture.

TOOLS & TECHNOLOGIES

Power BI

MySQL

METHODOLOGY

Data Collection

Data Preprocessing

Feature extraction

Log mining tasks

Results & insights

WHY IT MATTERS?

- **Early Threat Detection:** spots suspicious activity and security threats, before they escalate into serious attacks.
- **Faster Incident Response:** Instead of manually sifting through thousands of log lines, analysts get clear visual insights that cut response time significantly..
- **Preventing System Failures:** By predicting failures before they happen, it helps teams act proactively and keep systems running smoothly.

LOGS IN FOCUS

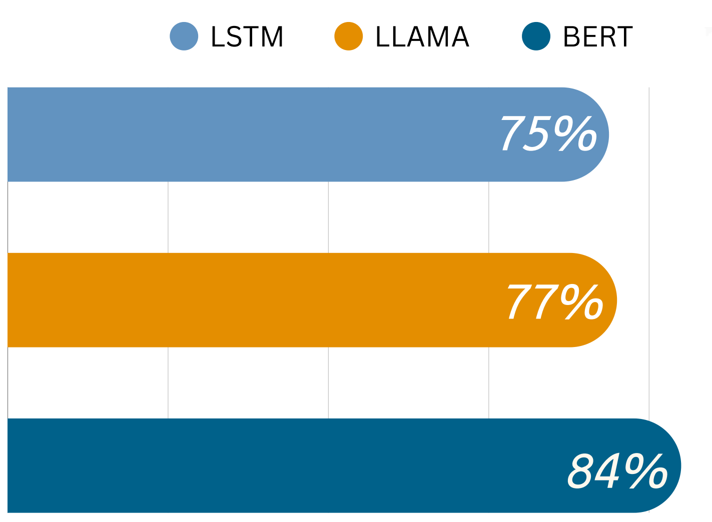
- Security Logs
- Network Logs
- Application Logs

KEY FINDINGS!

- Evaluated LSTM, LLaMA, and BERT models on the HDFS dataset.
- LSTM performed well but with slightly lower overall balance.
- System success is driven by a complete end-to-end pipeline.
- **The BERT-based model achieved the best performance, and after fine-tuning reached an F1-score of 0.92**, confirming its effectiveness for log analysis.

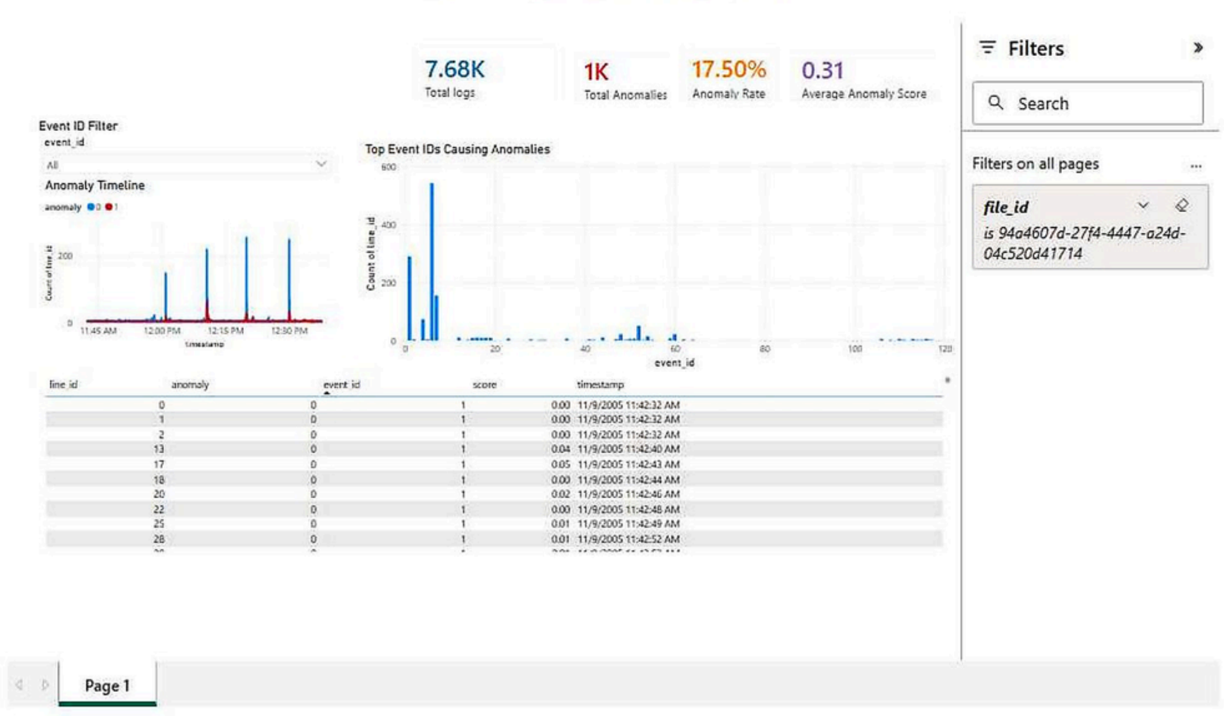
DATASETS USED

- HDFS 100k logs.
- BGL 100k logs.
- Thunderbird 100k logs.



"Models performance across the datasets."

ANALYSIS RESULT FOR TEST.CSV



"Analysis result taken from PowerBI."